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MILITARY SPECIFICATION

COMPRESSORS, RECIPROCATING, POWER-DRIVEN, OPEN-TYPE

FOR USE WITH REFRIGERANT-12

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers reciprocating, open-type compressors driven by prime movers with applied power of 10 horsepower (7460 Watts (W)) or less.

1.2 Classification. Open-type compressors shall be of the following types as specified (see 6.2).

Type I - Capacity group A and B of MIL-STD-279

Type II - Capacity group C and D of MIL-STD-279

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Command, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4130

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SPECIFICATIONS

FEDERAL

BB-F-1421	-	Fluorocarbon Refrigerants.
TT-E-516	-	Enamel, Lusterless, Quick-Drying Styrenated Alkyd Type.
TT-P-636	-	Primer Coating, Alkyd, Synthetic, Wood and Ferrous Metal.
TT-P-664	-	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
VV-L-825	-	Lubricating Oil, Refrigerant Compressor.
PPP-B-601	-	Boxes, Wood, Cleated-Plywood.
PPP-T-60	-	Tape: Packaging, Waterproof.

MILITARY

MIL-P-116	-	Preservation-packaging, Methods of.
MIL-P-514	-	Plates, Identification, Instruction and Marking, Blank.
MIL-S-901	-	Shock Tests, H.I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements For.

STANDARDS

FEDERAL

FED-STD-595	-	Colors.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-279	-	Compressor, Reciprocating, Power-Driven, Open-Type, For Use With Refrigerant-12.
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods.
MS 17248	-	Valves, Compressor, Service, Refrigerant-12.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

UNDERWRITERS LABORATORIES INC.

UL Standard No. 207(c) - Refrigerants Containing Components and Accessories, Nonelectrical

(Application for copies of this UL standard should be addressed to the Underwriters Laboratories Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, LI, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050; or 2602 Tampa East Boulevard, Tampa, FL 33619.)

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING ENGINEERS, INCORPORATED (ASHRAE)

Standard No. 23 - Methods of Testing for Rating Refrigerant Compressors

(Copies of ASHRAE publications may be obtained from the American Society of Heating, Refrigerating, and Air-Conditioning Engineers, Inc., 345 East 47th Street, New York, NY 10017.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Qualification. The compressors furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for the opening of bids (see 4.3 and 6.3).

* 3.2 Standard product. The compressors shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product with any added features needed to comply with the requirements of this specification. Modifications to add features shall not incorporate different parts unless such parts are used on other current commercial models. Standard or modified commercial products furnished in accordance with this specification shall be identifiable by all regular manufacturer's or commercial service organizations servicing the brand involved. Service organizations shall be capable of providing complete parts and repair services on models furnished to the Government consistant with their normal commercial practices.

* 3.3 Standards. The compressors shall comply with the applicable requirements of UL 207.

3.3.1 Compliance. Prior to approval of the first shipment of compressors the contractor shall submit to the contracting officer or his authorized representative satisfactory evidence that the compressors he proposes to furnish under this specification meet the applicable requirements of UL 207. Acceptable evidence of meeting the requirements of UL 207 shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the compressors have been tested and conform to UL 207.

3.4 Design. The design of the compressors shall conform to MIL-STD-279 and as specified herein. The compressors shall be reciprocating, open-type designed for operation with refrigerant conforming to type 12 of BB-F-1421. The compressors shall be capable of rotating in either direction. The design shall permit ready accessibility to all parts for maintenance, repair, and replacement. When specified, (see 6.2), the compressors shall withstand shock without damage (see 4.6.2.4). The complete compressor assembly shall contain no internal capacity control.

3.4.1 Storage temperatures. The compressors shall be capable of withstanding storage temperatures ranging from minus 65°F. (-54°C.) for at least 3 days duration to plus 155°F. (68°C.) for at least 4 hours without deterioration, leakage, entrance of moisture, or other impairment (see 4.6.1.1).

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3.4.2 Capacities.

3.4.2.1 Refrigeration capacities in Btu/hr. Compressors shall have the Btu/hr. capacities and horsepower inputs specified in MIL-STD-279 (see 4.6.1.2, 4.6.1.8, and 4.6.2.1). Capacities specified in MIL-STD-279 shall be met at rotative speeds not exceeding 80 percent of the maximum speed recommended by the compressor manufacturer. The unit shall be capable of repeating the capacity test results within ± 3 percent (see 4.6.1.2.2).

3.4.2.2 Pumping capacity. Compressors shall have a pumping capacity within 5 percent of the standard established in 4.6.2.1.2 when tested as specified in 4.6.2.1.3.

3.4.3 Adverse pressure operation. The compressors shall be capable of operating with a condensing temperature of 165°F. (74°C.) when operated for 8 hours (4 hours in each direction of rotation) at a speed required to meet the highest capacity group for the type, as specified in MIL-STD-279 (see 4.6.1.3).

3.4.4 Adverse starting. The compressor shall be capable of being started repeatedly with excess refrigerant in the crankcase oil. When tested as specified in 4.6.1.4 the compressor shall not slug, lose design oil pressure, stall, or have other mechanical malfunction.

3.4.5 Overspeed. The compressors shall be capable of operating for short periods at twice the speed required to meet the highest capacity group for the type, as specified in MIL-STD-279, without damage (see 4.6.1.5).

3.4.6 Operating temperature extremes. The compressors shall be capable of operating without damage or leaks under climatic conditions ranging from minus 40°F. (-40°C.) to 125°F. (52°C.) (see 4.6.1.6).

3.4.7 Endurance. The compressors shall be capable of operating continuously without damage, distortion, or excessive wear, for not less than 1000 hours at 125 percent of the highest speed required to meet the capacity group for the type as specified in MIL-STD-279 (see 4.6.1.7).

3.5 Materials. Where materials are not definitely specified, they shall be of the quality normally used by the manufacturer for open-type compressors provided the completed item complies with all provisions of this specification.

3.6 Construction. The construction of the compressors covered by this specification shall conform to MIL-STD-279 and as specified herein. The completely assembled compressors shall not leak refrigerant (see 4.6.2.2).

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3.6.1 Crankcase. The crankcase shall be fitted with an externally threaded hexagon head plug sized to provide a minimum clear opening of three-eighths of an inch diameter for adding oil. An oil drain hole with threaded hexagon head plug shall be provided at the bottom of the crankcase. A means for observing the oil level shall be provided either on both sides of the crankcase or on one side and on the end opposite the flywheel end.

3.6.1.1 Lubricating system. Compressors shall be lubricated by a positive forced feed system suppling oil to all main crankshaft and connecting rod bearings, and shaft seal housings. Oil shall be supplied by a pump directly connected to, or gear driven from the crankshaft of the compressor. The pump shall take oil from a sump fitted with an oil strainer. The oil pump shall operate in either direction without manual adjustment. The compressor shall be designed for and charged with oil conforming to type II of VV-L-825.

3.6.2 Flywheel. Unless otherwise specified (see 6.2), the compressors shall be equipped with a flywheel. Compressors with flywheels shall be statically and dynamically balanced.

3.6.3 Suction gas strainer. Compressors shall be fitted with a removable, internal suction gas strainer with not less than 40 mesh and shall not cause a pressure drop of more than 0.25 psi (17 kPa) (see 4.6.1.7.2).

3.6.4 Suction and discharge. The compressor suction and discharge valves shall be reed, disc, or ring types, assuring positive contact with the ground or finished seat. Valve action shall be positive over the complete design range of discharge and suction pressures.

3.6.5 Service valves. Compressors shall be equipped with suction and discharge service valves conforming to MS 17248 as follows:

	Type I compressor	Type II compressor
Suction valve	MS 17248-4	MS 17248-5
Discharge valve	MS 17248-3	MS 17248-4

3.6.6 Gage ports. In addition to the gage ports located on the service valves, 1/8 inch (3.175 mm) NPTF gage ports closed by a 1/8 inch (3.175 mm) NPTF plug shall be located on the suction and discharge sections of the compressor body.

3.6.7 Dehydration and holding charge. After the completion of all inspections each compressor shall be dehydrated and filled with oil specified in 3.6.1.1 and charged with (1) a holding charge of dry nitrogen sufficient to assure positive crankcase pressure of 5 to 10 psig (34 to 69 kPa) or (2) with a holding charge of dichlorodifluoromethane refrigerant. The moisture content of the oil and holding charge shall not exceed 25 parts per million (ppm) (see 4.6.2.3).

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3.6.8 Balancing. The compressor rotating parts shall be statically and dynamically balanced so that a vibration of 0.10 mil shall not be exceeded within the range of designed operating speed (see 4.6.1.7.2).

3.6.9 Seal. The compressor seal shall be designed so that the flexible member between the crankshaft and the carbon nose piece shall allow for an axial crankshaft movement of 0.030 inch minimum (0.076 mm).

3.7 Marking.

3.7.1 Warning tag. A tag 2 inches (51 mm) by 3 inches (76 mm) minimum shall be attached to each service valve and shall have a warning, printed in bold type, to the effect that the compressor is dehydrated and charged with nitrogen or refrigerant, as applicable, and oil, and that the valve is not to be opened until the compressor is ready for installation.

3.7.2 Indentification. Each compressor shall be marked for identification according to MIL-STD-130. The following information shall be placed on one or two plates. The plates shall conform to type III, composition A, class 1 or 2 of MIL-P-514 and shall be attached by rivets or screws.

(a) Nomenclature
(b) Manufacturer's name
(c) Manufacturer's model and serial number
(d) Date of manufacture
(e) Contract number
(f) Specification number
(g) Type number of compressor (see 1.2)
(h) "...rpm for capacity group..."
(i) "...rpm for capacity group..."
(j) "Use type II lubricating oil of VV-L-825"
(k) "Use Refrigerant 12"

3.8 Finish. Unless otherwise specified (see 6.2), cast-iron compressor bodies shall be primed with TT-P-636 and given a finish coat of enamel conforming to TT-E-516 with the color conforming to No. X24087 of FED-STD-595. In lieu of a painted finish, aluminum compressor bodies shall be anodized.

3.9 Workmanship. All parts, components, and assemblies of the compressor shall be clean and free from sand, dirt, fins, pits, sprues, scale, and flux. External surface shall be free from burrs, sharp edges, and corners. Bolts, nuts, screws, studs, and other types of threaded fasteners when used, shall not be broken, stripped, fractured or loose and shall have lockwashers, or shall be self-locking type.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. Qualification inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

* 4.3 Qualification inspection. When a qualification is required, the compressor shall be tested as specified in 4.6.1. The test specified in 4.6.1.8 shall be performed last. No order is specified for the other tests. One compressor of each type submitted shall be required to pass the qualification tests. Failure of any test shall be cause for failure of the compressor to qualify for listing.

* 4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.4.2 In-process inspection.

4.4.2.1 Visual examination of finish. Examination shall be made for application of finish specified in 3.8. Whenever nonconformance is noted, correction shall be made to the item affected and to the operation.

4.4.2.2 In-process dimensional examination. Components and subassemblies shall be examined for compliance with specification dimensional requirements. Any dimensions not within specified requirements shall be classified as a defect. The inspection level shall be S-1 with an acceptable quality level (AQL) of 4.0 defects expressed in terms of defects per hundred units.

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* 4.4.3 End item inspection. The lot shall consist of all compressors, of one type offered for inspection at one time. The sample unit for this examination shall be a completely assembled compressor.

4.4.3.1 Visual examination. The compressors shall be examined for the defects listed in table I. The inspection level shall be II with an AQL of 4.0 expressed in terms of defects per hundred units.

TABLE I. Visual defects

Examine	Defect
Construction and workmanship, general (applicable to all components and assemblies)	Fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, malformed, or otherwise impaired. Functioning component that required abnormal force to operate. Component missing or not type specified. Not connected or joined as specified.
Welding and brazing (where required)	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused.
Soldering (when applicable)	Missing, not adherent, or incomplete.
Finish	Color not as specified.
Bolts, nuts, screws, and other types of threaded fasteners	Missing, not type specified, broken, stripped, fractured or loose.
Marking identification and warning tag	Missing, incomplete, not legible, not specified type or size, not in proper location, not affixed in prescribed manner, not correct tag or plate.

4.4.3.2 Examination for weight and dimensions. The compressors shall be examined to determine compliance with weight and dimensional requirements. Weights or dimensions not within specified tolerances shall be classified a defect. The inspection level shall be S-2 with an AQL of 6.5 expressed in terms of defects per hundred units.

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4.4.3.3 End item testing. Each sample compressor shall be tested as specified in 4.6.2.1.2, 4.6.2.3, and 4.6.2.4, when applicable, using an inspection level of S-1 with an AQL of 4.0 expressed in terms of defects per hundred units. All compressors shall be tested as specified in 4.6.2.1.3 and 4.6.2.2 and any non-conformance shall be cause for rejection of the compressor and the lot.

4.4.4 Packaging inspection. An examination shall be made to determine that preservation-packing, and marking as required by section 5 are complied with. Defects shall be as specified below in table II. The sample unit shall be one shipping container fully prepared for delivery, except that it need not be sealed. The lot shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 expressed in terms of defects per hundred units.

TABLE II. Inspection of packaging

Examine	Defect
Markings (exterior container)	Missing; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any nonconforming component. Component missing, damaged or otherwise defective.
Preservation	Preservative improperly applied or missing.
Workmanship	Bulged or distorted container.

* 4.4.4.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-601, examination for defects in closure and strapping shall be in accordance with appendix of PPP-B-601. Examination for defects in cushioning, blocking and bracing, anchoring, and waterproofing shall be in accordance with MIL-STD-1186.

4.5 Code and standards compliance. Proof of compliance with applicable requirements of UL 207 shall be made available to the Government representative.

4.6 Test methods.

4.6.1 Qualification tests. Qualification tests shall consist of all tests in 4.6.1.1 thru 4.6.1.8.

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4.6.1.1 Storage test. Prior to any operating tests for qualification, the sample compressor with its normal oil charge and a minimum holding charge of 1 pound (0.45 kg) (avoirdupois) of refrigerant shall be stored in an ambient temperature of minus 65°F. (-54°C.) for 3 days, transferred to a room ambient temperature for a minimum of 24 hours and then stored in an ambient temperature of plus 155°F. (68°C.) for 4 hours. The compressor shall be examined for compliance with 3.4.1 during and after storage. Any nonconformance shall constitute failure of the test.

4.6.1.2 Capacity tests. Compressors shall be tested in accordance with ASHRAE Standard No. 23 for capacity, including horsepower inputs, to determine compliance with 3.4.2.1 as follows:

4.6.1.2.1 Capacity tests. Conditions for capacity tests covering both applicable groups shall be as follows:

(a) Ambient temperature: 110°F. (43°C.)

(b) Condensing temperatures: One run at 135°F. (57°C.), one run at 105°F. (40°C.)

(c) Suction temperatures: A run shall be made at each of the following sets of temperatures:

Saturated suction vapor temperature F.	Suction vapor temperature F.
-20 (-29°C.)	55 (13°C.)
-10 (-25°C.)	65 (18°C.)
0 (-18°C.)	65 (18°C.)
20 (-7°C.)	65 (18°C.)
50 (10°C.)	65 (18°C.)

Compressors shall be tested using the drive pulley diameters indicated in MIL-STD-279. Inability to meet capacities specified shall constitute failure of the test.

4.6.1.2.1.1 Extreme temperature capacity test. Each type compressor shall be tested for capacity of respective capacity groups at the following conditions:

(a) Ambient temperature: 125°F. (52°C.)

(b) Condensing temperature: 155°F. (68°C.)

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(c) Suction temperatures: A run shall be made at each of the following sets of temperatures:

Saturated suction vapor temperature °F.	Suction vapor temperature °F.
50 (10°C.)	65 (18°C.)
60 (16°C.)	70 (21°C.)

Inability to meet requirements shall constitute failure of the test.

4.6.1.2.2 Capacity verification test. A check shall be made for capacity of each respective capacity group of each type compressor at the minus 10°F. (-23°C.) saturated suction vapor temperature condition of 4.6.1.2.1. The resulting capacities using the primary method of 4.6.1.2.1 and making at 105°F. (41°C.) and 135°F. (57°C.) condensing temperature for each capacity group, shall be in accordance with 3.4.2.1. Inability to meet capacities shall constitute failure of test.

4.6.1.3 Adverse pressure test. The compressor shall be operated under the following adverse conditions to determine compliance with the requirements of 3.4.3. Inability to withstand operating pressures shall constitute failure of the test.

(a) Ambient temperature 135°F. (57°C.)

(b) Condensing temperature 165°F. (74°C.)

(c) Saturated suction vapor temperature 40°F. (4°C.)

(d) Suction vapor temperature 65°F. (18°C.)

4.6.1.4 Adverse starting test. The compressor shall be tested on a properly sized refrigerant circuit containing an oil separator and a suction pressure regulator. The regulator shall be set to maintain a maximum pressure of 19.2 psig (132 kPa) (-10°F. (-23°C.) saturated suction). Oil from separator shall be automatically returned to the compressor. The compressor in a stopped condition and charged with its normal amount of lubricating oil shall be permitted to come to temperature equilibrium at room ambient temperature. The suction valve shall be closed and the crankcase charged with refrigerant equal in weight to the weight of the lubricating oil. The suction valve shall then be opened and the compressor started immediately and operated at the speed required to meet the highest capacity requirements for the type, as specified in MIL-STD-279, at 135°F. (57°C.) condensing temperature for 30 minutes. The test shall be conducted 5 consecutive times. Inability to meet requirements of 3.4.4 shall constitute failure of the test.

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4.6.1.5 Overspeed test. The compressor shall be brought as quickly as possible from zero rpm to twice the speed required to meet the highest capacity group for the type, as specified in MIL-STD-279, with bypass and suction and discharge valves fully open and the source of power immediately cut off. This test shall be performed six times, three times in each direction of rotation, to determine compliance with 3.4.5. Inability to meet requirements shall constitute failure of the test.

4.6.1.6 High and low temperature test. The compressor shall be operated for 4 hours at an ambient of minus 40°F. (-40°C.) and for 4 hours at an ambient of 125°F. (52°C.) to determine compliance with 3.4.6. Inability to meet requirements shall constitute failure of the test.

4.6.1.7 Endurance test. The compressor shall be tested for endurance as follows to determine compliance with 3.4.7. Inability to meet requirements shall constitute failure of the test.

4.6.1.7.1 Conditions for endurance test. Conditions for the endurance test shall be as follows:

(a) Length of test: 1000 hours

(b) Speed: 125 percent of highest speed required to meet capacity for highest capacity group of type.

(c) Ambient temperature: 100°F. to 120°F. (38°C. to 49°C.)

(d) Condensing temperature: 135°F. (57°C.)

(e) Saturated suction vapor temperature: -10°F. (-23°C.)

(f) Suction vapor temperature: 65°F. (18°C.)

4.6.1.7.2 Procedure for endurance test. The compressor shall be operated continuously under conditions of 4.6.1.7.1 except for shutdowns necessary for reasons other than repair or adjustment of compressor. The following data shall be recorded:

Data	Method of recording
Ambient	Continuous recorder
Saturated suction vapor pressure	Continuous recorder
Suction vapor gas temperature	Continuous recorder
Condensing pressure	Continuous recorder
Horsepower input	Daily readings
Compressor rpm	Daily readings
Pressure drop	As required
Vibration of rotating parts	Daily readings
Nonoperating periods and reason therefor	As required

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The make, type, and serial number of the test instruments shall be included as a part of the test data furnished.

* 4.6.1.8 Examination after testing. After passing the storage, capacity, adverse pressure, adverse starting, overspeed, and endurance tests, the compressor shall be given a repeat test for capacity at the speed required to meet the highest capacity group for the type. The compressor shall then be completely disassembled and all parts examined for excessive wear, damage, and distortion. Horsepower requirements in excess of limits (see 3.4.2.1); a reduction of capacity in excess of 5 percent based on the average capacity obtained by the primary test method at minus 10°F. (-23°C.) saturated suction vapor temperature over the same average capacity obtained in the capacity test (see 4.6.1.2); or excessive wear, damage, or distortion shall be the cause of the compressor being rejected for qualification.

4.6.2 Quality conformance tests.

4.6.2.1 Capacity tests (check point and pumping).

4.6.2.1.1 Test conditions. The refrigeration capacity in Btu/hr. shall be determined at conditions of minus 20°F. (-29°C.) and plus 50°F. (10°C.) saturated refrigerant suction vapor temperatures and 135°F. (57°C.) condensing temperature.

4.6.2.1.2 Test procedures. The refrigeration capacity check test shall be performed by either a primary or confirming capacity test method of ASHRAE Standard No. 23 to determine compliance with 3.4.2.1. At the completion of the refrigeration capacity test, one of the compressors which has passed the test shall be operated on a calibrated flow test device normally used for compressor capacity flow testing. The compressor shall be run three times. The average flow rates shall be used as a standard to determine the acceptability of compressors tested in 4.6.2.1.3.

4.6.2.1.3 Pumping capacity. Compressors shall be on the same flow measuring device used in 4.6.2.1.2. Compressors shall be operated at the same speed as compressors used as standard in 4.6.2.1.2. Failure to comply with pumping capacity within 5 percent of standard will constitute failure of test.

4.6.2.2 Final leak test. Following the capacity tests of 4.6.2.1, all compressors shall be dehydrated and subjected to an internal refrigerant gas pressure of 50 psig (345 kPa) minimum. The compressor shall then be thoroughly leak tested with a halide leak detector or an electronic leak detector at room temperature to determine compliance with 3.6. Any indication of a leak shall constitute failure of the test.

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4.6.2.3 Moisture content test. Moisture content test shall be performed on compressors having received the oil and holding charge of nitrogen or refrigerant specified in 3.6.7 after the compressor has remained in an ambient temperature of not less than 90°F. (32°C.) for a period of at least 3 hours. Moisture content shall be determined by means of a moisture indicating sight glass or by any other commercially accepted method of equal sensitivity to determine compliance with 3.6.7. Moisture in excess of 25 ppm shall constitute failure of the test.

4.6.2.4 Hi-shock test. The compressors shall be subjected to a shock test conforming to type A of MIL-S-901 to determine compliance with 3.4. Inability to meet requirements shall constitute failure of the test.

5. PACKAGING

5.1 Preservation-packaging. Preservation-packaging shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. All exposed, uncoated ferrous metal surfaces of each compressor shall be cleaned process C-1 of MIL-P-116 and thoroughly dried. Surfaces, cleaned as specified, except the pulleys, shall be coated with type P-7 preservative of MIL-P-116. Exposed surfaces of the pulleys shall be coated with primer conforming to TT-P-664. All openings into the compressor shall be sealed with tape conforming to type I or II, class 1 of PPP-T-60.

5.1.2 Commercial. Each compressor shall be preserved to afford adequate protection against corrosion and deterioration during shipment from the supply source to the first receiving activity. The contractor may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

* 5.2.1 Level A packing. Each compressor, preserved as specified in 5.1, shall be packed in a shipping container conforming to overseas type, style A or B, type 3 load of PPP-B-601. Cushioning, blocking and bracing, anchoring and waterproofing shall be in accordance with MIL-STD-1186. Each shipping container exceeding 250 pounds gross weight shall be provided with skids in accordance with PPP-B-601. Closure and strapping shall be in accordance with the appendix of PPP-B-601.

5.2.2 Level B packing. Each compressor, preserved as specified in 5.1, shall be packed as specified in 5.1.1, except that the shipping container shall conform to domestic type, style A or B, type 3 load of PPP-B-601 and waterproofing shall not be required.

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5.2.3 Commercial packing. Each compressor, preserved as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The Military standard open-type compressors covered by this specification are intended for use in combination with other Military standard refrigeration components in refrigeration and air-conditioning systems.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type of compressor required (see 1.2).
(c) When high shock test is required (see 3.4).
(d) When flywheel is not required (see 3.6.2).
(e) When finish is other than specified (see 3.8).
(f) Selection of applicable levels of preservation-packaging and packing (see 5.1 and 5.2).

6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the U.S. Army Natick Research and Development Command, Natick, MA 01760 and information pertaining to qualification of products may be obtained from that activity.

6.3.1 If any design change to a qualified product is made or contemplated, the design data shall be submitted to the responsible activity prior to the bid opening date for evaluation, testing, and requalification if it is deemed necessary.

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6.4 Changes from previous issue. The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

Project No. 4130-0243

Review activities:

Army - CE, ME

User activities:

Navy - CG, MC, SH

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